Fuzzy Sets

- Introduced by Lotfi A Zadeh in 1960's
- Used to represent sets where boundary of information is unclear
- To account for concepts used in human reasoning which are vague and imprecise
- Elements can belong to the set or not
- Strength of membership/ Degree of membership is associated

Example

- Fuzzy set is very convenient method for representing some form of uncertainty
- For example: the weather today
  - Sunny: If we define any cloud cover of 25% or less is sunny
  - This means that a cloud cover of 26% is not sunny?
  - Vagueness should be introduced

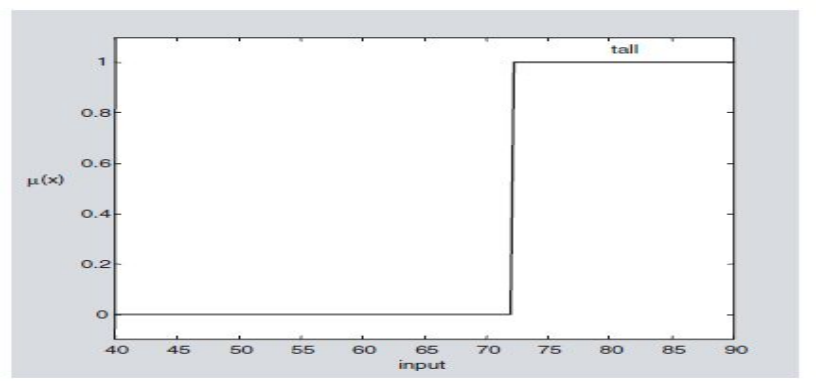
Difference

- Ordinary Sets-Only two values possible
- Membership of element 'x' in set A is described by a characteristic function $\mu_A(x)$ which can be either 0 or 1
- Fuzzy sets - Extends using partial membership
- A fuzzy set A on a universe of discourse U is characterized by a membership function $\mu_A(x)$
  that takes values in the interval [0, 1]

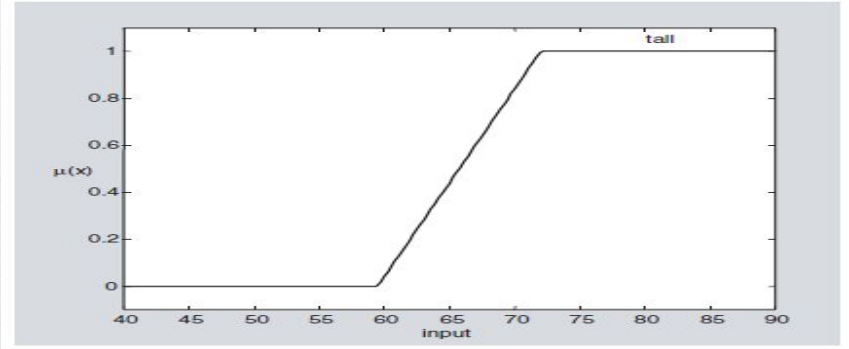
Fuzzy Example - Tall

- A fuzzy set A in U may be represented as a set of ordered pairs. Each pair consists of a generic element x and its grade of membership function; that is

$$A = \{(x, \mu_A(x)) \mid x \in U\}$$



Ordinary Set



Fuzzy Set

Fuzzy Membership Functions

- One of the key issues in all fuzzy sets is how to determine fuzzy membership functions
- A membership function provides a measure of the degree of similarity of an element to a fuzzy set
- Membership functions can take any form, but there are some common examples that appear in real applications

Fuzzy sets- subset

- Given two fuzzy set A,B defined on the Universe of Discourse X, then *A* is a subset of *B* denoted by

$$A \subseteq B$$

- Iff $\mu_A(x) \leq \mu_B(x)$ for all $x \in X$ $x \in U$

$A \subseteq B$ iff $A \cap B = A$ and $A \cup B = B$ for any $A, B \in \mathcal{F}(X)$.

Fuzzy Complement

- To be able to develop fuzzy systems we also have to deal with NOT or complement.
- This is the same in fuzzy logic as for Boolean logic
- For a fuzzy set *A*, *A* denotes the fuzzy complement of *A*
- Membership function for fuzzy complement is

$$\mu_{A}(x) = 1 - \mu_{A}(x)$$

Fuzzy Intersection

- Most commonly adopted t-norm is the minimum
- Given two fuzzy sets A and B with membership functions $\mu_A(x)$ and $\mu_B(x)$, the intersection A and B defined over the same universe of discourse X is a new fuzzy set A∩B also on X with membership function which is the minimum of the grades of membership function of every x to A and B

$$\mu_{A\cap B}(x) = \min(\mu_A(x), \mu_B(x))$$

Fuzzy Union

- Given two fuzzy sets A and B with membership functions $\mu_A(x)$ and $\mu_B(x)$, the union A and B defined over the same universe of discourse X is a new fuzzy set $A \cup B$ also on X with membership function which is the maximum of the grades of membership function of every x to A and B
- $\mu_{A \cup B}(x) \equiv \mu_A(x) \vee \mu_B(x) \equiv \max(\mu_A(x), \mu_B(x))$

Example Problem 1

Let $U = \{ 1,2,3,4,5,6,7\}$

$A = \{ (3, 0.7), (5, 1), (6, 0.8) \}$ and

$B = \{(3, 0.9), (4, 1), (6, 0.6) \}$

Find $A \cap B$, $A \cup B$, $B-A$ and A'

$A \cap B = \{ (3, 0.7), (6, 0.6) \}$

$A \cup B = \{ (3, 0.9), (4, 1), (5, 1), (6, 0.8) \}$

$A' = \{(1, 1),(2, 1), (3, 0.3), (4, 1), (6, 0.2),(7, 1)\}$

$B-A = \{ (3, 0.3), (4, 1), (6, 0.2)\}$

Fuzzy Logic Laws

- **Intersection distributive over union...**

$$\mu_{A\cap(B\cup C)}(x) = \mu_{(A\cap B)\cup(A\cap C)}(x)$$

min[A,max(B,C)]=max[min(A,B), min(A,C)]

- **Union distributive over intersection...**

max[A,min(B,C)]= min[max(A,B), max(A,C)]

$$\mu_{A\cup(B\cap C)}(x) = \mu_{(A\cup B)\cap(A\cup C)}(x)$$

Fuzzy Logic Laws

- **Obeys Demorgan's Laws**

$$u_{\overline{(A \cap B)}}(x) = u_{\overline{A} \cup \overline{B}}(x)$$

$$u_{\overline{(A \cap B)}}(x) = u_{\overline{A} \cup \overline{B}}(x)$$

Fuzzy Logic Laws Contd..

- Fails The Law Contradiction

$$A \cap \overline{A} \neq \phi$$

- Thus, (the set of numbers *close* to 2) AND (the set of numbers <u>not</u> *close* to 2) ≠ null set

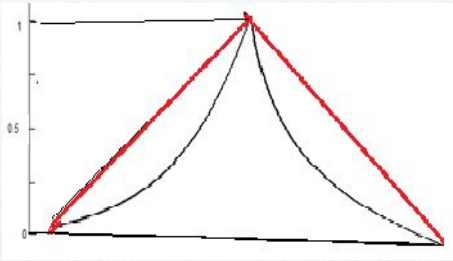
Other Results

- $\neq X$
- $= X$

Basic Operations

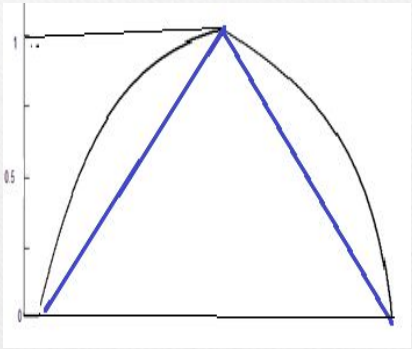
- For reshaping the membership functions
  - **Dilation (DIL) :** increases the degree of membership of all members by spreading out the curve
  - **Concentration (CON):** Decreases the degree of membership of all members
  - **Normalization (NORM)** : discriminates all membership degree in the same order unless maximum value of any member is 1. Computed as: $\mu A(x) / \max(\mu A(x)), x \in X$

Graphical representation



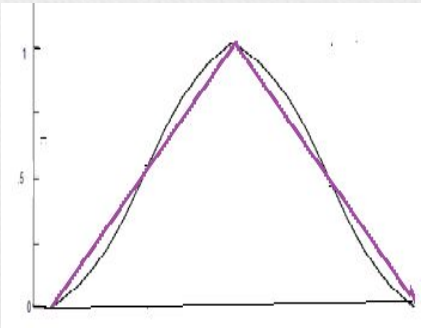
- Concentration

  $CON(A) = [u_A(x)]^2$



- Dilation

$DIL(A) = [u_A(x)]^{1/2}$



- Intensification

Reasoning with Fuzzy Logic

- Premise A
- Implication relation R(x,y)
- Conclusion B’
- Fuzzy value A’ matches approximately with A

Inference Procedure

Modus Ponens:

Premise:		*x* is A'
Implication:	R(x,y)	if *x* is A then *y* is B
Consequence:		*y* is B'

Where A, A', B, B' are fuzzy sets and *x* and *y* are symbolic names for objects

Example

- Premise : This banana is very yellow
- Implication : If a banana is yellow then the banana is ripe
- Conclusion : This banana is very ripe

Inference

- $R_c(y)=R_A(x) \circ R_B(x,y)$ where $\circ$ signifies the composition of A & B
- Commonly used method for composition

is Max-Min

- $R_c(y)=\max_x \min \{u_A(x), u_B(x,y)\}$

Inference Example

X=Y={1,2,3,4}

A={little}={(1/1),(2/0.6),(3/0.2),(4/0)}

R=approximately equal, in fuzzy relation defined by

	y	1	2	3	4
1		1	0.5	0	0
2		0.5	1	0.5	0
3		0	0.5	1	0.5
4		0	0	0.5	1

Inference Example contd..

$R_c(y)=\max_x \min \{u_A(x), u_R(x,y)\}$

$= \max_x \{\min [(1,1),(0.6,0.5),(0.2,0), (0,0)] ,$

$\min [(1,0.5),(0.6,1),(0.2,0.5), (0,0)]$

$\min [(1,0),(0.6,0.5),(0.2,1), (0,0.5)]$

$\min [(1,0),(0.6,0),(0.2,0.5), (0,1)] \}$

$= \max_x \{[1,0.5,0,0],[0.5,0.6,0.2,0],[0,0.5,0.2,0],[0,0,0.2,0]\}$

$= \{ [1],[0.6],[0.5],[0.2] \}$

Inference Example contd..

Therefore the solution is

$R_c(y)=\{(1/1),(2/0.6),(3/0.5),(4/0.2)\}$

Started in terms of fuzzy modus ponens we might interpret this inference

Premise : x is little

Implication : x and y are approximately equal

Conclusion : y is more or less equal

Generalisation

The before mentioned notions can be generalized to any number of universals by taking the cartesian product and defining the various subsets

Thank You